TrES-2b

R-0137

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_TRES-2_180604` · created 2026-07-16T06:50:02+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458273.79605 +/- 0.00054 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1178 +/- 0.0087
Transit depth (Rp/Rs)^2	1.39 +/- 0.2 [%]
Orbital Inclination (inc)	84.68 +/- 0.41
Airmass coefficient 1 (a1)	1.1963 +/- 0.0078
Airmass coefficient 2 (a2)	-0.1573 +/- 0.0053
Scatter in the residuals of the lightcurve fit is	0.66 %
Best Comparison Star	None
Optimal Aperture	4.68
Optimal Annulus	6.14
Transit Duration (day)	0.0842 +/- 0.005

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1178 ± 0.0087 vs 0.1254 (deviation 0.85σ)
Duration fit vs published	0.0842 d vs 0.0746 d (deviation 1.88σ)
Mid-time O–C	-1.5 min
Depth SNR	13.2
Residual scatter	0.66 %

Light curve

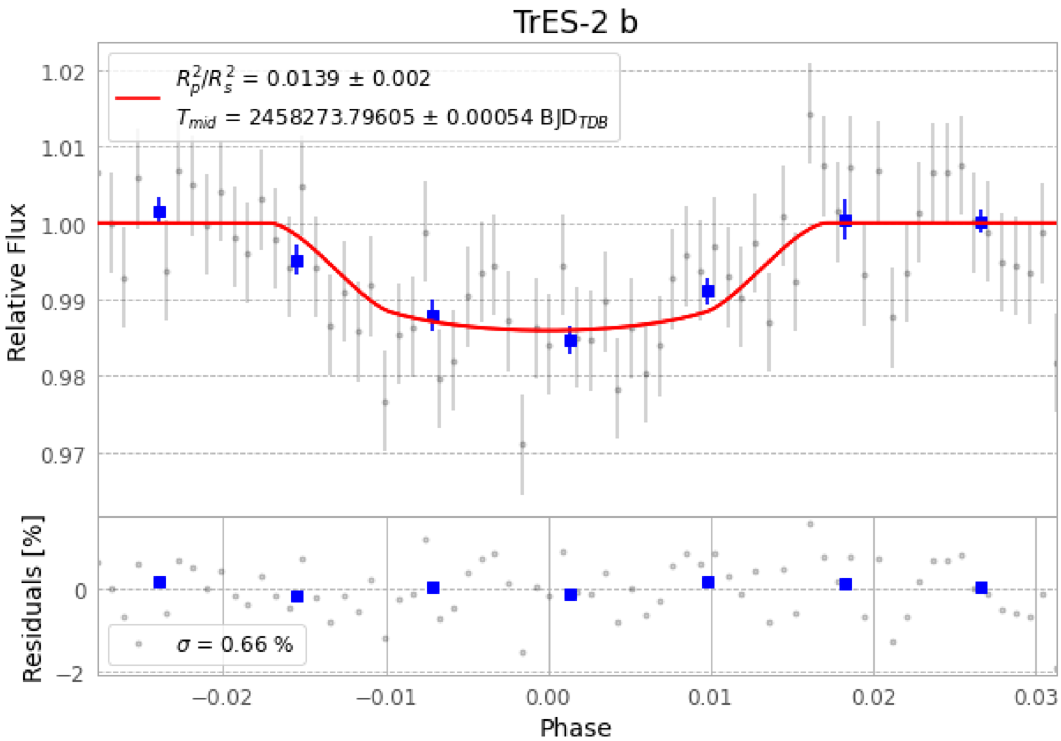

Figure 1 — FinalLightCurve_TrES-2 b_2018-06-03.png (SHA-256 5b63b38981a6928b…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [408, 221], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 0931b8ff5693f5eb…).
`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 3ebc8ba`

Data provenance

71 FITS frames (47 MB), `TRES-2180604052412.FITS` → `TRES-2180604085412.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-2-180604.log (de6390c4d5a2…), FinalLightCurve_TrES-2 b_2018-06-03.png (5b63b38981a6…), FinalLightCurve_TrES-2 b_2018-06-03.csv (de9c8e9bed18…)

Compute resources

Wall time 115.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)